

Machine Data Learning

Python ML Yuxi Hayden Liu

Artificial Intelligence: A Modern Approach
Stuart Russell and
Peter Norvig

AI:- That which mimics human intelligence in machines.
The science and engineering of making machines (like comp. robots) intelligent ~~and~~ enabling them to take actions and decisions that humans do.

Machine:- Physical system that uses power to apply forces and control movements, perform actions (to make life easier).

Complex machines are combination of 2 or more simple machines.

AI has been proliferating in healthcare etc.

AI for social good:- Computer Vision, help farmers, supply chain management, Public & wild life safety, Automated Event Discovery, Border Patrol, etc.

What is intelligence?

The ability to understand, learn and think.

The capacity- Wiki defn preps.

We think, learn from observations, and make decisions.

But what about machines?

Why is he such an ass? :-

Two dimensions to AI

Humanness vs Rationality

Act vs Thinking

Turing Test Human interpreter cannot distinguish b/w comps & humans after processing posing some written questions
C needs NLP, Knowledge rep, Automated reasoning, ML, CV & Robotics

Turing Test involves interaction with objects and people

~~Think~~ Think Humanly :- Cognitive modelling

Act Rationally :- Agent based modelling, Agents should operate autonomously, perceive the environment, adapt to change

Think Rationally :- Right thinking, Logic based approach, need to involve probability to model uncertain things

What is ML?

Construction of algos that can learn from and make predictions on data.

data == features (feature modelling?) [CL2 saving my nose lol]

Supervised ML :- Classification, Regression [Bayesian modelling]

Unsupervised ML :- Clustering, Anomaly detection

features for differentiating b/w cricket and tennis ball would be wt and diameter [plot on graph, CL vectors]

Perceptron:- Observe feats, apply activation fn and produce output

Linearity in features makes computation easier

If $x_1 = \text{width}$ and $x_2 = \text{wt}$

$$f_n = w_0 + w_1 x_1 + w_2 x_2$$

↓

bias $\rightarrow$ need to distinguish b/w the 2 balls easily

Neural Networks and Deep Learning

Replicate neurons in human brain

4

Where does AI stand today?

Significant progress in last three decades in CV, NLP, speech recog.

Far from ~~tricking~~ mimicking humans perfectly

Human intelligence is not purely ~~and~~ reflex mechanisms but is dependent on reasoning that operates on the internal representations of knowledge

AI is about building machines and sys that can reason. learn & solve problems similar to humans. logical reasoning is key to allowing sys to decide based on given conditions & facts

Knowledge base :- Set of sentences rep'd in knowledge representation language.

↳ Expressing knowledge explicitly in a computer-tractable way. (Logic? Not?)

Logic :- A way of thinking or explaining, sensible reasons for doing smthg, sci of thinking about or explaining the reasons for smth using formal methods.

Basis of all mathematical and automated reasoning

Formal vs Informal Logic

↓

Formal Lang

Study of logical Truth

Symbolic / Mathematical Logic

Natural Lang

Study of informal fallacies, crit. thinking

Natural Lang exhibit ambiguity

Makes ~~reasoning~~ reasoning difficult laniguity

Formal Lang

promotes rigour, reduces possibility of human error

helps reduce implicit / unstated assumptions by removing familiarity with subject matter

helps achieve generality due to possibility of finding alt interpretation for sentences & args.

Formal Logic:- Uses syllogisms to make inferences & examines how conclusions follow from premises based on the structure of args

Mathematical Logic:- Uses math symbols to prove theoretical args

Symbolic Logic:- Uses symbols to accurately represent valid & invalid args

Propositional Logic

Imp. cause of AI being

① Core of AI:- Possibility of automating reasoning & drawing inferences from knowledge

Answer queries

discover facts that follow from the knowledge base, decide what to do

Essential for knowledge rep, reasoning, decision-making process

Entailment:- Deducing a statement from other statements

$$T \rightarrow T$$

$$F \rightarrow T/F$$

$$T/F \leftarrow T$$

$$F \leftarrow F$$

① $\forall$ Humans have 2 eyes

② Sujit is human

③ Sujit has 2 eyes

④ $\forall$ Humans have 4 eyes

⑤ ~~②~~ Sujit is human

⑥ Sujit has 4 eyes

Pack my bag with five dozen liquor jugs

Pack my bag with five dozen liquor jugs

Fallacy:- Conclusion may be right, but reasoning is faulty

Propositional Logic:- Deals with propositions (statements) which are T/F. Propositional calculus. Zero-order logic. Foundation of higher order logics

Order of Logic Degree of quantification that can be performed over sets.

- (i) 1st order:- Quantifies only over individuals. Predicate logic / calculus
- (ii) 2nd order:- Over sets / predicates (acc to Prats :3)
- (iii) 3rd order:- Over sets of sets
- (iv) Higher orders:- Quantification over arbitrarily deep nesting

Propositions are declarative statements

- (i) Atomic:- Simple, indivisible, cannot be chunked further. Represents basic fact or condition.
"The door is closed"
- (ii) Compound:- Atomic props combined using logical connectives (AND, OR, NOT)
"The door is closed AND the heater is on"

Connectors:-

- (i) AND ($\wedge$)
- (ii) OR ($\vee$)
- (iii) NOT ($\neg$)
- (iv) IMPLIES ($\rightarrow$) $S = T$ if $P \rightarrow Q$
- (v) IFF ($\leftrightarrow$) $P \leftrightarrow Q = \neg P \text{ XOR } Q = P \rightarrow Q \wedge Q \rightarrow P$

$P \rightarrow Q$:- P is necessary for Q
Q is sufficient for P

$P \leftrightarrow Q$:- P & Q are equivalent
 Q is necessary & sufficient

Truth Table :- break down of all possible truth values returned by a logical expr.

P	Q	$P \rightarrow Q$	$\neg P \vee Q$	$P \leftrightarrow Q$	$\neg P \oplus Q$
0	0	1	1	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	1	1	1	1	1

P & Q can be compound propositions

Exercise:- A & $B = T$, x & $y = F$

- (i) $\neg(A \vee x)$ F
- (ii) $A \vee (x \wedge y)$ T
- (iii) $A \wedge (x \vee (B \wedge y))$
- (iv) $((A \wedge x) \vee \neg B) \wedge \neg((A \vee x) \vee \neg B)$
- (v) $(x \wedge y) \wedge (\neg A \vee x)$
- (vi) $((x \wedge y) \rightarrow A) \rightarrow (x \rightarrow (y \rightarrow A))$

Terminology

→ Valid sentences (Null sentences)

Tautology:- Invariably True for all assignments

SAT:- A sentence is satisfiable iff there is some assignment of values that results in True

A sentence is UNSAT if it is $\neg$ SAT

Towards Formal Proof

$\alpha \Rightarrow \beta$ when all the formulae in set α are True, β is True

Semantic notion, concerns the notion of truth (??)
(Grammar died)

Construct a Truth Table for α, β

$\alpha \Rightarrow \beta$ if $\exists$ row where all formulae in α are true, β is also True

Modus Ponens:-

$\alpha: P \wedge (P \rightarrow Q); \beta: Q$

Write TT for $P, P \rightarrow Q$

Therefore $P, P \rightarrow Q \Rightarrow Q$

Intends to formally capture the notion of proof that is commonly applied in other fields

Proof:- of a formula from a set of premises is a seq of steps in which any step of the proof is

i) an axiom

ii) a formula deduced from the prev step using some rule of inference
(b)

The last step of the proof should deduce the formula we wish to prove.

We say that S follows from (premises) α to denote that the
Towards Formal

set of formulae α "proves" the formula S

Soundness:- A logic is sound if it preserves truth
(i.e. if a set of premises are all true, any conclusion drawn from those premises must also be true)

Completeness:-

- (i) Semantic: A logic is complete if it is capable of proving any valid consequence
- (ii) Syntactic: Completeness requires that every sentence, or its negation is provable (i.e. a theorem)

Decidability:- A logic is decidable if $\exists$ a mechanical procedure (computer program) to prove any given consequence.
(or) $\exists$ a decision procedure that is finite for theorem

No logical system is syntactically complete

Zero Order Logic is semantically complete, so is FOL
ZOL is decidable, FOL is not

ZOL inference rules

- (i) Modus Ponens $P, P \rightarrow Q \Rightarrow Q$
- (ii) Modus Tollens (Contraposition) $P \rightarrow Q, \neg Q \Rightarrow \neg P$
- (iii) Hypothetical Syllogism $P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$
- (iv) And Elimination $\bigwedge_{i=1}^n P_i \Rightarrow P_i$
- (v) And Introduction $P_1, P_2, \dots, P_n \Rightarrow \bigwedge_{i=1}^n P_i$
- (vi) Or Introduction $P_i \Rightarrow \bigvee_{i=1}^n P_i$
- (vii) Not Double Negation Elimination $\neg \neg P \Rightarrow P$
- (viii) Unit Resolution $P \vee Q, \neg Q \Rightarrow P$
- (ix) Resolution $P \vee Q, \neg Q \vee R \Rightarrow P \vee R$

Example:-

α

β

- 1) $A \vee (B \rightarrow D)$
- 2) $\neg C \rightarrow (D \rightarrow E)$
- 3) $A \rightarrow C$
- 4) $\neg C$

$$B \rightarrow E$$

Proof

- ⑤ $D \rightarrow E$ 2, 4 (Modus Ponens)
- ⑥ $\neg A$ 3, 4 (Modus Tollens)
- ⑦ $B \rightarrow D$ 1, 5 (Unit Resolution)
- ⑧ $B \rightarrow E$ 5, 7 (Hypothetical Syllogism)

Q) ~~Construct formal proof for:-~~
~~If the investigation~~

$$\begin{array}{l} S \rightarrow G \\ \neg S \rightarrow E \\ \neg G \rightarrow E \end{array}$$

Conjunctive Normal Form (CNF)

A literal is a propositional letter or the negation of a propositional letter

A clause is a disjunction of literals

CNF $\rightarrow$ conjunction of clauses

DNF $\rightarrow$ Disjunction of ~~clauses~~ conjunction of literals

Every propositional logic-formula can be converted to CNF and DNF

Converting to CNF

- (i) Eliminate $\leftrightarrow$ rewriting $P \leftrightarrow Q$ as $(P \rightarrow Q) \wedge (Q \rightarrow P)$
- (ii) Eliminate $\rightarrow$ as $P \rightarrow Q \equiv \neg P \vee Q$
- (iii) Use ~~push~~ De Morgan's laws to push $\neg$ inwards
- (iv) Rewrite $\neg(P \wedge Q)$ as $\neg P \vee \neg Q$
- (v) Rewrite $\neg(P \vee Q)$ as $\neg P \wedge \neg Q$
- (vi) Eliminate double negation $\neg \neg P \equiv P$
- (vii) Use distributive laws to get CNF
- (viii) Rewrite $(P \wedge Q) \vee R$ as $(P \vee R) \wedge (Q \vee R)$
- (ix) Rewrite $(P \vee Q) \wedge R$ as $(P \wedge R) \vee (Q \wedge R)$

Q) Convert $\neg(P \rightarrow (Q \wedge R))$

$$\begin{aligned} & \neg(P \rightarrow (Q \wedge R)) \\ & \neg \neg P \wedge \neg(Q \wedge R) \\ & P \wedge (\neg Q \vee \neg R) \\ & \Phi \end{aligned}$$

Barchuri Time! (Hopefully don't regret this)

Bro's sharing slides! Goat!

Only taking down topic names, ought to refer to the textbooks

ML :- Scientific study of algos & statistical models that computer systems use

1) to perform a specific task effectively sans explicit instr.

4) rely on patterns & inferences instead

Involves building a math model based on sample data

$$\begin{aligned}
 E[(y - \hat{f}(x))^2] &\Rightarrow E[(f + \epsilon - \hat{f})^2] = E[(f - \hat{f})^2 + \epsilon^2 + 2\epsilon(f - \hat{f})] \\
 &= E[(f - \hat{f})^2] + E[\epsilon^2] + 2E[\epsilon(f - \hat{f})] \\
 &= E[(f - E[\hat{f}] + E[\hat{f}] - \hat{f})^2] + \sigma^2 + 0 \quad [\text{Independence of noise}] \\
 &= E[(f - E[\hat{f}])^2 + (E[\hat{f}] - \hat{f})^2 + 2(f - E[\hat{f}])(E[\hat{f}] - \hat{f})] + \sigma^2 \\
 &= E[(f - E[\hat{f}])^2] + E[(E[\hat{f}] - \hat{f})^2] + 2(f - E[\hat{f}])E[E[\hat{f}] - \hat{f}] + \sigma^2 \\
 &= (f - E[\hat{f}])^2 + \text{Var} + 0 + \sigma^2 = \text{Bias}^2 + \text{Var} + \sigma^2 \quad //
 \end{aligned}$$

Q) Let $\hat{f}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$
 Minimise MSE $\rightarrow$ no of training samples

$$J(\theta) = \frac{1}{n} \times \min \sum_{i=1}^m (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

We wish to penalise x^2 & x^3

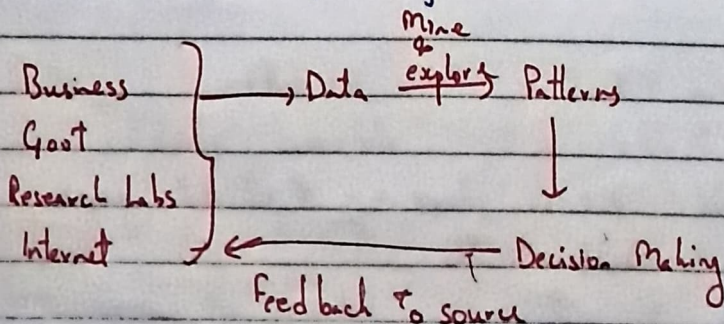
Add $+1000 \theta_2 + 1000 \theta_3$

Back To Sujit!

Data Science:-

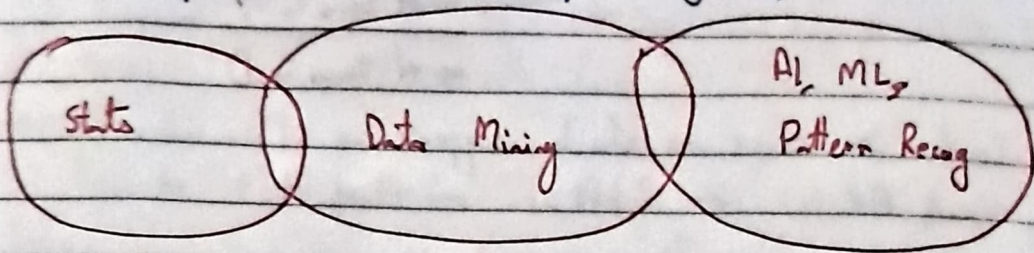
$\hookrightarrow$ clean $\rightarrow$ visualise (or visualise acc to Sujit) $\rightarrow$ analyse
 predict $\leftarrow$

Interdisciplinary academic field that uses stats, scientific computing, scientific methods, processing, ... (ref wiki for defn)



Data Mining is the process of discovering valuable insights, patterns and information from vast datasets by employing various techniques, algorithms & tools

This is a technology that blends traditional data analysis methods with sophisticated algs for processing large amounts of data.



Data base Tech, llnd comp, Distr. Comp

Patterns:-

- ↳ Association:- Coffee buyers also buy sugar.
- ↳ Clustering:- Segregation & segmentation of data.
- ↳ Classification:- Bank customer applies for loan, Will they default?

Association Rules:-

↳ Antecedent & Consequent

↳ AR: $X \rightarrow Y$

$\{x_1, \dots, x_n\} \rightarrow \{y_1, \dots, y_m\}$

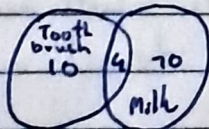
↳ itemset is the list of all the items in the antecedent & the consequent

i.e. $X \cup Y$

Assoc. Rule

$$\text{Support} = \frac{4}{100}$$

For ex:-



A.R:- Toothbrush $\rightarrow$ Milk

$$\text{Conf} = \frac{4}{14}$$

$$\text{Support} \Leftarrow \frac{\text{no. of transactions in } X \cap Y}{\text{no. of transactions in } D} = P(X \cap Y)$$

$$\text{Confidence} \Leftarrow \frac{\text{no. of transactions in } X \cap Y}{\text{no. of transactions in } X}$$

Ex:- Let $D = 200$

① $X = \text{Sugar} ; |X| = 100 ; Y = \text{Milk} ; |Y| = 160 ; |X \cap Y| = 70$

Support = 0.35 ; Conf = 0.7

② $X = \text{Coffee} ; |X| = 50 ; |X \cap Y| = 38$

Support = 0.19 ; Conf = 0.76

③ $X = \text{Bread} ; |X| = 20 ; Y = \text{Butter} ; |Y| = 20 ; |X \cap Y| = 15$

Support = 0.075 ; Conf = 0.75

Lift is symmetric across X & Y

↑

apsara

Date: _____

$$\text{Lift}(AR) = \frac{P(Y|X)}{P(Y)} = \frac{\text{no of trans in } X \cap Y}{\frac{\text{no of trans in } X}{\text{no of trans in } D}} = \frac{|X \cap Y| \times |D|}{|X| \times |Y|}$$

(conf ≥ 0.05 $\rightarrow$ check support $\rightarrow$ if considerable
good AR $\leftarrow$ if lift > 1 $\leftarrow$ check lift $\leftarrow$

$$X \rightarrow Y \rightarrow \text{no of AR} = \sum_{k=1}^n {}^n C_k \times (2^{n-k} - 1)$$

$$X \subseteq [n]$$

$$Y \subseteq [n]$$

$$= 3^n - 2^{n+1} + 1 \quad \text{By inclusion exclusion}$$

$\hookrightarrow$ each element has 3 choices.

But we still have to explore 2^{101} itemsets if they meet reasonable support

$$|X \cup Y| < \text{threshold} \rightarrow |X \cup Y \cup Z| < \text{threshold}$$

If any set has a bad support, all its supersets also have a bad support.

$\hookrightarrow$ A priori algorithm.

Itemset can be frequent only if all its subsets are frequent

No superset can be frequent if the itemset is not frequent

Example:

tx.id		Trial 1	Trial 2	Trial 3
1	{1, 2, 3, 4}	1 = 3	1, 2 = 3	
2	{1, 2, 4}	2 = 5	1, 3 = 1	
3	{1, 2}	3 = 4	1, 4 = 2	
4	{2, 3, 4}	4 = 5	2, 3 = 3	
5	{2, 3}		2, 4 = 1	
6	{3, 4}		3, 4 = 3	
7	{2, 4}			

Pseudo code is:

- 1) Create 1-sized frequent itemset lists that meet a threshold support
- 2) Expand itemsets list by combining overlapping sets to size $k+1$
- 3) Remove infrequent itemsets from the list
- Repeat
- 3) Prune the expanded lists using the apriori property
- loop 2, 3, 4 until no expansion is possible

Algorithm:-

Apriori (T, ϵ):

$L_1 = \{\text{large singleton itemsets}\}$

$k = 2$

while L_{k-1} is not empty:

$C_k = \text{generate_candidates}(L_{k-1}, k)$

for t in T : if $c \subseteq t$

$D_t = \{c \text{ for } c \text{ in } C_k \text{ for } c \text{ in } C_k\}$

for c in D_t

count[c] = count[c] + 1

$L_k = \{c \text{ if count}[c] \geq \epsilon \text{ for } c \text{ in } C_k\}$

$k = k + 1$

return Union $\{L_k\}$ for all k

generate_candidates(L, k):

result = ~~empty~~ set()

for all p, q in zip(L, L):

if p and q differ by one element exactly:

$c = p \cup q$

for $u \subseteq c$:

if $u \in L$ and $|u| = k-1$:

result.add(c)

return result

Hierarchical and Cyclic association rules also exist

Eclat Algorithm:- Eq. class transformation, a DFS strategy. Works best when the dataset is dense.

F-P Growth:- Freq. Pattern Growth, a compact data structure. FP-tree to compress the data set. Works better than the rest when the data is dense and large.

Classification:-

We collect different measurements / facts / features

$$x = \{x_1, \dots, x_d\}$$

set of classes $y = \{1, 2, \dots, k\} = [k]$ (set of k)

If $k=2 \leftarrow$ binary classification
else $\leftarrow$ multiclass

How do we measure the goodness of the classifier?

Accuracy: $P(\hat{y}_x = y_x)$
 $\nearrow$ gold
 $\hookrightarrow$ predicted value

$$= \frac{\text{no of pts classified correctly}}{\text{no of data pts}} = \frac{TP + TN}{TP + TN + FP + FN}$$

But fails in case of skewed data.

So we assign loss wts to fix this issue.

But we usually use Recall & Precision

True $\Rightarrow$	P	TP	FN
	N	FP	TN

Recall = True Positive Rate = $\frac{TP}{TP+FN}$

Precision = Positive Prediction Value = $\frac{TP}{TP+FP}$

False Negative Rate = $\frac{FN}{TP+FN}$

FPR = $\frac{FP}{FN+TN}$

TNR = $\frac{TN}{TN+FP}$

F₁ Score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$

Bayes Classifier :-

$Y \in \{1, \dots, k\}$; Prior = $P(Y=k)$ = probability that a data point belongs to class k ; $P(Y=k | X=x)$ = Posterior . We assign the MAP estimate: $\text{Max}_{k \in \{1, \dots, k\}} P(Y=k | X=x) = \frac{P(X=x | Y=k) \times P(Y=k)}{P(X=x)}$.
 A Posteriori = $\arg \max_k P(Y=k | X=x)$. Likelihood $\times$ Prior
 Naive Bayes Classifier :- Assumption of $P(X=x) \rightarrow$ evidence or normalizing const

Conditional Independence . Optimal Classifier on accuracy = error rate is minimised .
 But learning the likelihood is very hard, hence we do not apply this in 1x instead we use naive Bayes .

Decision Tree :- How do we decide what the P_k is? (create a decision tree maximising information gain at each step) ID3 $\rightarrow$ Heuristic Dichotomiser, Ross Quinlan in 1980s

Entropy :- 1 bit = 2 combinations, 2 bits = 4 combinations, etc. Formula = $-\sum_i P_i \log_2(P_i)$
 $H(S) = -\sum_{i \in [k]} \frac{x_i}{s} \times \log_2\left(\frac{x_i}{s}\right)$. Information Gain = $H(S) - \sum_i \frac{x_i}{s} \times H(S_i)$

CART = Classification and Regression Tree $\rightarrow$ Only 2 children $\rightarrow$ Some goodness criterion to split (minimise Gini index) $[\sum p_i(1-p_i)] \rightarrow$ Pruning to avoid overfitting. Decision Trees

are very sensitive to noise in data . k Nearest Neighbours . $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$. Given a point x , find out the k nearest neighbours (nearest points from A to x)

$\hat{y}$ = most frequently occurring label $x_{(1)}, x_{(2)}, \dots, x_{(k)}$. How to choose the right value of k ? Estimate errors for different k values . Choose k with least error

Runtime = linear in $n \times d$. As $n \rightarrow \infty$, error rate = $2 \times$ Bayes error .
 $d: X \times X \rightarrow \mathbb{R}$. (i) $d(x, x) = 0$ (ii) $d(x, y) = 0$ iff $x = y$; (iii) $d(x, y) + d(y, z) \geq d(x, z)$

Freq. classes dominate : $L_p = \sum_{i=1}^n \|x_i' - x_i''\|_p^{1/p}$. Combining classifiers :- Bagging : Take majority vote for the best class in each new record . Boosting : Each classifier's vote has a wt proportional to its acc on training data

Clustering

- ↳ Getting things simple people or objects
- ↳ Spatial Clustering.

Hard Clustering:- Each point is assigned to some cluster

Soft Clustering:- Each point is assigned a probability of being assigned to each cluster

We wish to partition data such that $\forall i, j; i \neq j; S_i \cap S_j = \emptyset$
and $\bigcup_{i=1} S_i = \{x_1, \dots, x_n\}$

Connectivity based Clustering:-

(i) Agglomerative (AGNES):-

- ↳ def each point as a cluster
- ↳ group nearby clusters
- ↳ merge until # clusters = defined value

(ii) Divisive (DIARMA):-

- ↳ All points in one cluster
- ↳ Remove extreme points
- ↳ Regroup removed pts.
- ↳ Repeat until # clusters = defined value.

Nominal / Categorical Variables

$d(x, y) = 1 - m/n$ where m = no of matches among n attributes
 m = sum of wts of matching attributes & n = sum of wts of all attributes.

Dist w.r.t. a cluster could be centroid, avg dist, min dist, etc.
Context dependent

* Within a cluster variance is minimised

If k is given, simply minimise the variance of each cluster

$$\sum_{i=1}^k \sum_{x \in S_i} \|x - M_i\|^2$$

$$M_i = \text{centroid} = \frac{1}{|S_i|} \sum_{x \in S_i} x$$

$$= \min_{i=1}^k \sum_{|S_i|} \sum_{x \in S_i} \|x - y\|^2$$

But all these formulas are NP hard.

k-means:- Randomly select k pts as centres, or
Assign each point to the cluster closest to these centres.
Compute mean of all the pts in the cluster
All these pts the new centres of the clusters

What is the best 'k'?

Project on 2/3D and determine visually

Silhouette Score:-

$SC(i)$ = Based on avg dist with pts in the same cluster $[a(i)]$ vs smallest distance to the points not in cluster $[b(i)]$.

$$SC(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}, \text{ Avg}(SC(i)), \text{ Higher better.}$$

Davies - Bouldin Index

Captures compactness of a cluster with its separation across clusters

$$DB = \frac{1}{k} \sum_i \max_j \left(\frac{\Delta x_i + \Delta x_j}{\delta(x_i, x_j)} \right)$$

Δ = intra cluster distance, δ = intercluster distance

Best k minimizes DB

MDP - decides what action to be taken at states
MC - decisions are made, just find the value
Endhu

క 2 చిత్

గతమునకు వీరి

కన్య చంద్రులకు కన్య చంద్రులకు వీరి

అంతా చిన్మయమునకు వీరి

అంతా చిన్మయమునకు వీరి

క 2 చిత్

గతమునకు వీరి

State space = $\{(H,H), (H,L), (L,H), (L,L)\}$

$P(H|H) = 0.6$

$P(H|L) = 0.6$

$P(L|H) = 0.4$

$P(L|L) = 0.4$

Find best action at each step & perform backward induction

Reinforcement Learning

Q-learning, SARSA, SARSA^(A), DDPG

Q? $|S| = 1$

$$S = \{s\} \quad A = \{1, 2, \dots, k\}$$

$$R(s, a) = \begin{cases} 1, & \text{if } a = a^* \\ 0, & \text{otherwise} \end{cases}$$

$$V(s, a) = R(s, a) + \gamma(V(s, a))$$

$$= \frac{R(s, a)}{1 - \gamma} = \frac{\mu_a}{1 - \gamma}$$

$$V(s) = \max_a \frac{\mu_a}{1 - \gamma}$$

But what if the probabilities are not known?

Multi Arm Bandits.

↳ action

Classical problem in decision making under uncertainty

Key Elements:-

↳ K arms (actions/choices)

↳ Each arm has a stochastic reward distribution, parameters unknown

↳ At each round, select one arm & observe reward

Goal is to maximise cumulative reward over T rounds

Balance b/w exploration & exploitation

Learn rewards while optimising

Greedy Strategy:-

Assume Bernoulli Trials

Estimate expected reward $\hat{\mu}_i$ for each arm

At every time t , select the arm with the highest $\hat{\mu}_i$

$$\hat{\mu}_a^{t+1} = \begin{cases} \frac{\hat{\mu}_a^t \times N_{a,t} + 1}{N_{a,t} + 1} & \text{if } R = 1 \\ \frac{\hat{\mu}_a^t \times N_{a,t}}{N_{a,t} + 1} & \text{otherwise} \end{cases}$$

Purely exploitation based, Greedy.

Round Robin:- Purely exploration

Greedy Algo

Input: K arms

Initialisation: Pull each arm $i \in [K]$ once to initialize $\hat{\mu}_i$

for T in $\{K+1, \dots, T\}$:

select $i^* \in \arg\max_{i \in [K]} \hat{\mu}_{i,t}$

observe reward $r_{i^*,t}$

update empirical mean $\hat{\mu}_{i^*}$ using $r_{i^*,t}$

ϵ -Greedy

Gives a simple method for exploration.

At each time step

Exploitation: $P = \epsilon \rightarrow$ Randomly choose an arm

Exploitation: $P = 1 - \epsilon \rightarrow$ Greedy Action

ϵ -Greedy Algo

Input = K, ϵ ; Initialisation = Pull each arm once to init $\hat{\mu}_i, N_i$

for t in range $(K+1, T+1)$:

sample p from $U[0,1]$

if $p < \epsilon$:

select arm i^* from $U[1, K]$

else:

select arm $i_t \in \underset{j \in [K]}{\operatorname{argmax}} \hat{\mu}_{j,t-1}$

Observe reward $r_{i_t,t}$

Update count $N_{i_t}(t) \leftarrow N_{i_t}(t-1) + 1$

Update $\hat{\mu}_{i_t,t} \leftarrow \hat{\mu}_{i_t,t-1} + \frac{1}{N_{i_t}(t)} (r_{i_t,t} - \hat{\mu}_{i_t,t-1})$

Perf measure

Need to quantify how well an algorithm performs
Understand the cost of learning

Regret :-

Measure the stochastic difference b/w optimal and realised / achieved performance.

Capture the loss due to not knowing the best arm initially
Lower regret = better algorithm

Expected Regret

Let $r_{i_t,t}$ denote the reward observed by pulling arms according to an algorithm A

$$\operatorname{Reg}_A^{\text{exp}}(T) = E \max_{1 \dots K} \left[\sum_{t=1}^T r_{i^*,t} - r_{i_t,t} \right]$$

But this is very hard to compute / bound, hence we use a slightly weaker notion that is easier to bound. That is called pseudo-expected regret

Pseudo-regret

$$\operatorname{Reg}_A^{\text{pseudo}}(T) = \sum_{t=1}^T \max_{1 \dots K} E[r_{i_t}] - E \left[\sum_{t=1}^T r_{i_t,t} \right] = \sum_{t=1}^T E[\mu_{i^*} - \mu_{A_t}]$$

$$\operatorname{Reg}_A(T) = T \times \mu_{i^*} - \sum_{t=1}^T \mu_{i_t}$$

~~Expected Pseudo Regret~~

We do not fix ϵ in ϵ -Greedy.

One soln is to set $\epsilon = \frac{1}{t}$ where t is time

But do we need this much exploration.

UCB:-

Using confidence bounds to guide exploration

- ↳ Explore arms that could be optimal
- ↳ Naturally reduce exploration as confidence increases.

UCB - Upper Confidence Bound

Select the arm with the highest upper confidence bound.

$$\mu_i \in (\hat{\mu}_i - c_i, \hat{\mu}_i + c_i)$$

$$c_i = \sqrt{\frac{\alpha \ln T}{N_i(t)}}$$

α is a hyperparameter

Algorithm for UCB

Input $\alpha > 0$

for $t \in \{1, \dots, K\}$ do

select an arm $i_t \leftarrow t$

observe the reward $r_{i_t, t}$

set initial $M_i \leftarrow r_{i_t, t}$

end for

for $t \in \{K+1, \dots, T\}$ do

select arm $i_t \in \arg \max_{i \in [K]} \hat{M}_{i, t} + \sqrt{\frac{\alpha \ln T}{N_i(t)}}$

observe reward $r_{i_t, t}$

update $\hat{M}_{i_t}$

end for

UCB Regret

$$\text{Reg}_{\text{UCB}}(T) \leq \left[8 \sum_{i: M_i < M^*} \frac{\ln T}{\Delta_i} \right] + \left(1 + \frac{\pi^2}{3} \right) \left(\sum_{j=1}^K \Delta_j \right)$$

Where $\Delta_i = \mu^* - M_i$ denotes the suboptimality gap of arm i

Set $\alpha = 2$

Key Takeaway - Regret grows logarithmically with time T
i.e. $O(\ln T)$

Problem proposed in 1951, proof in 2002

We might not know Δ_i because we don't know μ^*

Instance independent $\rightarrow \text{Regret} = O(\sqrt{T})$

" ~~Instance~~ independent $\rightarrow \forall \Delta_i \rightarrow$ formula above

Recomposition:-

Goal :- Bound the expected no. of times a suboptimal arm i is pulled $\mathbb{E}[N_i(T)]$

At any time t , if a suboptimal arm i is chosen over the optimal arm i^* , it implies that

$$\hat{\mu}_{i,t}^+ \geq \hat{\mu}_{i^*,t}^+ \quad \left[\begin{array}{l} \text{The } + \text{ shows the upper bound,} \\ - \text{ for lower bound} \end{array} \right]$$

This error occurs if only one of 3 events happens

We use confidence radius

$$c_{t,s} = \sqrt{\frac{2 \ln t}{s}}$$

Event 1:- Underestimation of the optimal

$$\hat{\mu}_{i^*,t} \leq \mu^* - c_{t,s}$$

Event 2:- Overestimation of the suboptimal

$$\hat{\mu}_{i,t} \geq \mu_i + c_{t,s}$$

Event 3:- High Uncertainty - The confidence interval is still wide enough to cover the gap Δ :

$$\mu^* < \mu_i + 2c_{t,s}$$

Bounding Probabilities:-

Bounding events 1 & 2 (concentration) :- Using Chernoff

- Hoeffding's bounds, the probability of the empirical mean deviating by $c_{t,s}$ decays rapidly:

$$P(\text{Event 1}) \leq t^{-4} \text{ and } P(\text{Event 2}) \leq t^{-4}$$

Summing these over all t results in a small regret.

$$\sum_{t=1}^{\infty} 2t^{-4} \rightarrow \frac{\pi^2}{3}$$

These events contribute only $O(1)$ to the expected regret

Bounding Event 3 (Deterministic Limit)

Event 3 implies that we haven't pulled arm i enough times yet

$$\sqrt{\frac{2 \ln T}{N_{i,t}}} > \frac{\Delta_i}{2} \Rightarrow N_{i,t} < \frac{8 \ln T}{\Delta_i^2}$$

Once arm i is sampled $I = \frac{8 \ln T}{\Delta_i^2}$ times, the conf interval

shrinks enough that this event cannot occur.

Conclusion:- The regret is dominated by the samples needed to shrink the interval, giving the $O(\ln T)$ bound:

$$E[N_i(T)] \leq \frac{8 \ln T}{\Delta_i^2} + \left(4 + \frac{\pi^2}{3}\right)$$

$$\text{Regret is just } \mathbb{E}[N_i(T)] \times \sum_{j=1}^K \Delta_j$$

no of rounds
from arm j

no of times
pulled

Side Note :- Sample $M_i \sim \Gamma(\Delta_i, B_i)$

Results in Regret in $O(\ln T)$

Thompson Thompson Sampling $\Rightarrow$ Not in syllabus

Variations of MAB

Stochastic

Adversarial

Stationary

Non stationary

Volatile / Sleeping Bandits

Budgeted Bandits

2)

Given

$$U_{t+1}(I) = \max_A [C(I, A) + \gamma \sum_j \Pi_j]$$